Analyzing Smart City Bike Sharing Data using Power BI

Problem Statement:

Public bike-sharing systems generate continuous data from hundreds of stations across different cities. The task is to analyze this real-time bike station dataset to understand station performance, usage efficiency, and operational patterns across cities.

Applying Power BI skills—from data cleaning and modeling to DAX and dashboarding—to turn raw data into actionable urban mobility insights.

Dataset

Source Description : Dataset link

Tools & Technologies

• **Power BI:** Data Cleaning, Dimension Table, Data modelling, DAX calculations, visualization, and interactive dashboard creation.

Data Pre-Processing Power Query)

Data Entry & Import

The Bike data set was imported into Microsoft Power BI using the Get Data option from the Excel/CSV source. This enabled structured loading of raw data into the Power BI environment for further transformation and modeling.

Data Cleaning & Transformation

1. Column Renaming

- Some column names in the dataset were unclear or inconsistent.
- To improve readability and usability, columns were renamed with clear and meaningful names.Original column names were modified into descriptive names such as

Original Column Name	Renamed Column Name	Reason
Number	Station ID	Represents the unique identifier for each bike station
Name	Station Name	Indicates the name of the bike station
Address	Station Address	Shows the location/address of the station
Position	Location Coordinates	Contains lat long of the station

Banking	Banking Facility	Indicates whether bank payments facility is available
Bonus	Bonus Availability	Shows if bonus service is available at the station
Status	Station Status	Indicates whether the station is open or closed
Contract Name	City	Represents the city where the station is located
Bike Stands	Total Bike Capacity	Shows the total no of stand avvailable for bikes
Available Bike Stands	Available Bike Stands	Indicates How many stands are currently empty
Available Bike	Available Bikes	Shows the number of bike currently available
Last updated	Last Updated time	Indicates the last time the station data was updated.

2. Removed Duplicates

- Using Power Query, duplicates were removed by selecting the Station ID column and applying the Remove Duplicates function.

3 Handling Missing Values

- Some records contained null values in the Station Address column.
- Null values were replaced with the text "Unknown".
- To avoid blank or missing values in the dataset.
- To maintain data completeness.

4 Changing Data Types

- Correct data types were assigned to ensure accurate calculations and aggregations
- Station ID, Statiion Name, address, Banking, Bonus, City were changed to data types as Text
- Last Updated Time column was converted to Date/time types as date.
- Non Numerical column was converted to whole Number

5 Split position column

The dataset contained a **Position column** that stored geographic coordinates.

The column was split using a **delimiter** to extract:

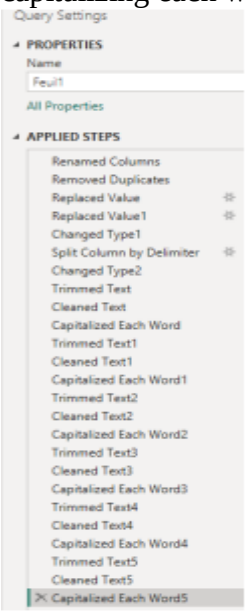
- Latitude
- Longitude

- To enable map-based visualizations.
- To improve geographic analysis of bike stations.

6. Standardizing Text Format

Applied **TRIM** – Removed extra leading and trailing spaces

CLEAN – Removed hidden or non-printable characters
Text fields such as **Station Name and Station Address** were standardized by capitalizing each word.



7. Calculated Column

Occupied Bike Stands
Shows how many stands are currently occupied by bikes.
Occupied Stands = 'Fact_Bike Station'[Total Bike Capacity]-'Fact_Bike Station'[Available Bike Stands]

Bike Availability %
Shows percentage of bikes available at each station.
Change **data type → Decimal Number**.
Bike Availability = DIVIDE('Fact_Bike Station'[Available Bikes],'Fact_Bike Station'[Total Bike Capacity],0)*100

Station Utilization

Station Utilization measures how much of a station's total capacity is currently being used. It is calculated by dividing the number of occupied stands by the total stand capacity of the station.

Station Utilization % = DIVIDE([Occcupied Stands],[Total Bike Capacity],0)*100

Station Load Level

Station Load Level categorizes stations based on their utilization percentage. This helps identify whether a station has low, medium, or high usage.

Station Load Level = SWITCH(TRUE(),'Fact_Bike Station'[Available Bikes]<='Fact_Bike Station'[Total Bike Capacity]*0.2,"Low Bikes(High Demand)",'Fact_Bike Station'[Available Bikes]<='Fact_Bike Station'[Total Bike Capacity]*0.5,"Medium Bikes","High bikes(Low Demand)")

Empty Station

Empty Station identifies stations where no bikes are available. This column flags stations that currently have **zero available bikes**. These stations are considered empty and indicate areas where demand for bikes is higher than supply.

Empty Station = IF('Fact_Bike Station'[Available Bikes]=0,1,0)

Occupied Stands	Bike Availability	Station Utilization %	Station Load Level	Empty Station
4	5	20	Low Bikes(High Demand)	0
0	0	0	Low Bikes(High Demand)	1
1	0	6.66666666666667	Low Bikes(High Demand)	1
0	0	0	Low Bikes(High Demand)	1
1	8.33333333333333	8.33333333333333	Low Bikes(High Demand)	0
2	10	10	Low Bikes(High Demand)	0
23	92	92	High bikes(Low Demand)	0
0	0	0	Low Bikes(High Demand)	1
19	95	95	High bikes(Low Demand)	0
2	16.6666666666667	16.6666666666667	Low Bikes(High Demand)	0
6	25	30	Medium Bikes	0
0	0	0	Low Bikes(High Demand)	1

Creation of Dimension Table

Fact Table

main dataset becomes the Fact Table. Renamed to**Fact Bike Station**

Date Dimension Dim_Date

A Fact Table was used to create the Dim_Date table using calender function.
A continuous date range was generated to cover the entire analysis period. Duplicate dates were eliminated. Date data type was correctly assigned.

```
DimDate =
ADDCOLUMNS (
    CALENDAR ( MIN ( 'Fact_Bike Station'[Date] ), MAX ( 'Fact_Bike Station'[Date] ) ),
    "Year", YEAR ( [Date] ),
    "Month Number", MONTH ( [Date] ),
    "Month Name", FORMAT ( [Date], "MMMM" ),
    "Day", DAY ( [Date] )
)
```

Date	Year	Month Number	Month Name	Day
01-01-2024 00:00:00	2024	1	January	1
02-01-2024 00:00:00	2024	1	January	2
03-01-2024 00:00:00	2024	1	January	3
04-01-2024 00:00:00	2024	1	January	4
05-01-2024 00:00:00	2024	1	January	5
06-01-2024 00:00:00	2024	1	January	6
07-01-2024 00:00:00	2024	1	January	7
08-01-2024 00:00:00	2024	1	January	8
09-01-2024 00:00:00	2024	1	January	9
10-01-2024 00:00:00	2024	1	January	10
11-01-2024 00:00:00	2024	1	January	11
12-01-2024 00:00:00	2024	1	January	12
13-01-2024 00:00:00	2024	1	January	13
14-01-2024 00:00:00	2024	1	January	14
15-01-2024 00:00:00	2024	1	January	15

Dim_Station

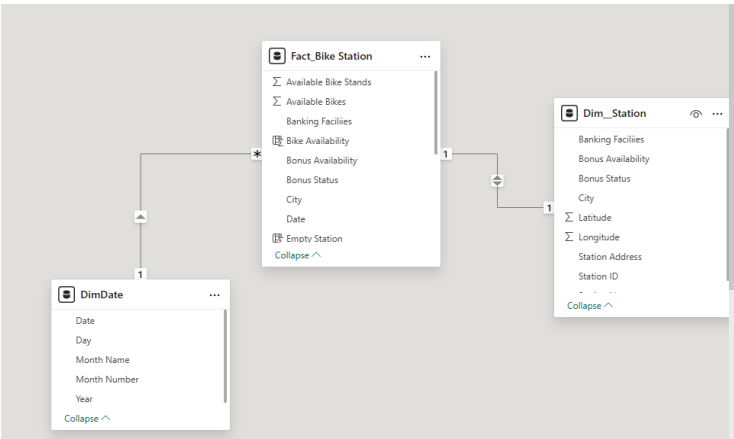
Stores unique station information.
A reference of the Fact Table was used to create the Dim_Station table.

Station ID	Station Name
3068	3068 - Faure / Meynis
17003	17003 - St Fons - Theatre
3033	3033 - Woodstower 2025
3040	3040 - Faure / Danton
8055	8055 - Saint-Romain / Berliet
6017	6017 - Lafayette / Inkermann
5051	5051 - Lugdunum - Radisson
22001	22001 - Ste Foy - Place St Luc
24001	24001 - Ecully - Mairie
4008	4008 - Clos Savaron
11005	11005 - Caluire - Passerelle De La Paix
23001	23001 - Tassin - Tuileries
2027	2027 - Perrache Est
19001	19001 - St Genis L. - Médiathèque
8033	8033 - Berthelot / Cazeneuve
5032	5032 - Ménival / Eisenhower
10045	10045 - Alsace Anatole France
26001	26001 - St Didier

The dataset was cleaned, standardized, and structured using proper data transformation techniques including null handling, duplicate removal, text cleaning, and data type correction. A calculated Price Category column was created at the transaction level to enable price segmentation analysis, ensuring an optimized and analytically data model

Final Data Model Preparation

- A star schema data model was created by connecting the Fact Table with
- Dim_Date, Dim_Station, Dim_Date ausing one-to-many
- relationships.
- The Fact Table contains Bike data while the dimension tables store
- descriptive information such as date and station
- This data model improves data organization, enables efficient filtering, and
- supports accurate analysis and visualization in Power BI dashboards.
- Below shows the relationship diagram as fact table in the centre connecting to all
- dimension table.



Dax Measures

Total Available Bikes

Total Availability Bikes = SUM('Fact_Bike Station'[Available Bikes])

Total Bike Capacity
Total Bike Capacity = SUM('Fact_Bike Station'[Total Bike Capacity])

Total Available Bike Stand
Total Availability Bike Stands = SUM('Fact_Bike Station'[Available Bike Stands])

Bike Utilization
Bike Utilization = DIVIDE(SUM('Fact_Bike Station'[Available Bikes]),SUM('Fact_Bike Station'[Total Bike Capacity]),0)

Occupied Stand
Occcupied Stands = SUM('Fact_Bike Station'[Total Bike Capacity])-SUM('Fact_Bike Station'[Available Bike Stands])

Total Station
Total Station = DISTINCTCOUNT('Fact_Bike Station'[Station Name])

Bike Availability
Bike Availability % =
```
DIVIDE(
    'Measures Table'[Total Availability Bikes],
    'Measures Table'[Total Availability Bike Stands]
) * 100
```

Station Utilization
Station Utilization % = DIVIDE([Occcupied Stands],[Total Bike Capacity],0)*100

Empty Station Count
Empty Station Count =
```
CALCULATE(
    COUNTROWS('Fact_Bike Station'),
    'Fact_Bike Station'[Available Bikes]= 0
)
```

Average Station Utilization
Average Station Utilization = AVERAGE('Fact_Bike Station'[Station Utilization %])

Gauge Maximum
Gauge Maximum = 100

Gauge Minimum Gauge Minimum = 0

Target Value
Target Value = 70

Analysis and Visualizations (Power BI)

KPI OVERIEW

A total of 4 KPI includes using card visuals

Total Station

This KPI shows the **total number of bike stations available in the network**. It represents the overall infrastructure of the bike sharing system across the city.

Occupied Stands

This KPI represents the **number of stands currently occupied by bikes**. It indicates how many bikes are currently parked across all stations.

Average Bike Availability

This KPI shows the **average number of bikes available across all stations**. It helps understand the overall availability of bikes for users.

Station Utilization %

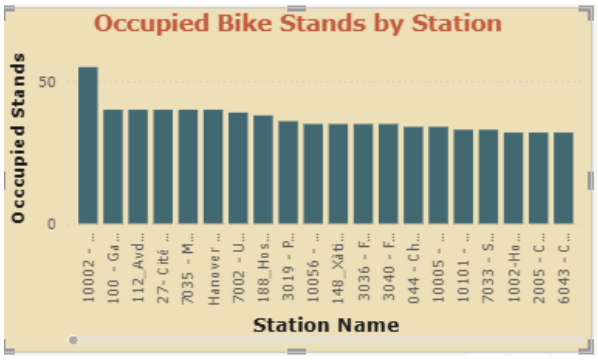
This KPI measures the **percentage of station capacity currently being used**. It indicates how efficiently the bike stands are utilized across the network.

1028	8937	28.77	46.02
Total Station	Occcupied Stands	Average Bike Availability	Station Utilization %

Chart Analysis

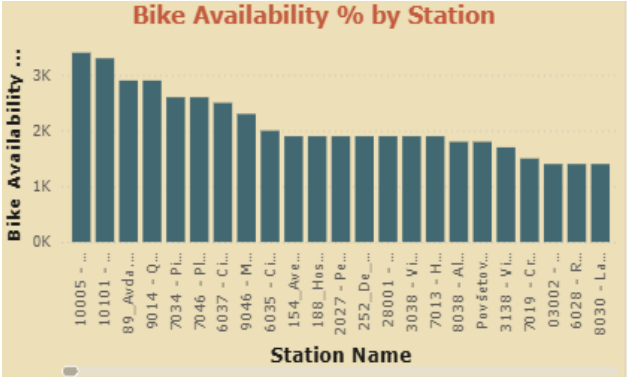
Occupied Bike Stands by Station

Which bike stations are used the most?



- Some stations show **higher occupied stands**, meaning more bikes are currently parked there.
- A few stations have **lower occupied stands**, indicating lower bike availability or higher bike usage.
- Stations with high occupied stands may have **lower demand**, while stations with fewer occupied stands may indicate **higher usage by customers**.

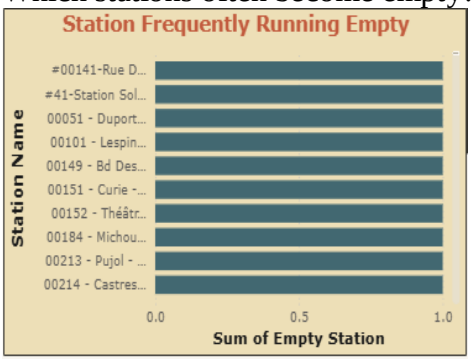
Bike Availability % by Station



- Bike availability percentage varies significantly between stations.
- Some stations have **very high bike availability**, meaning many bikes are ready for use.
- Other stations have **lower availability**, suggesting frequent rentals.
- Stations with low availability may require **better bike rebalancing to meet demand**.

Station Frequently Runnung Empty

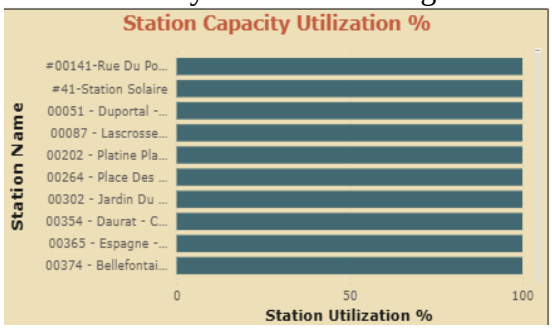
Which stations often become empty?



- Several stations appear in the **empty station list**, meaning they occasionally run out of bikes.
- This indicates **high customer demand at these locations**.

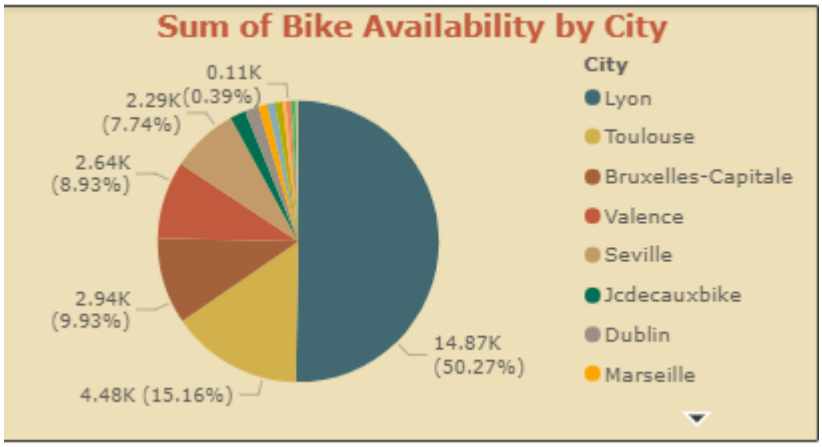
Station Capacity utilization %

How efficiently are stations being used?



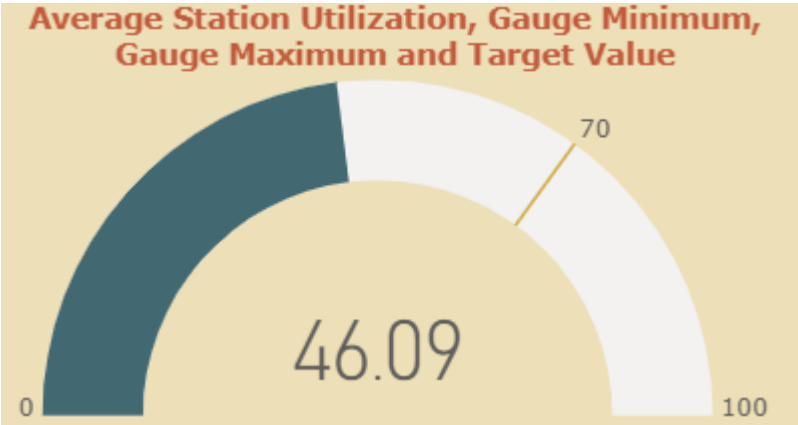
- Station utilization percentages show how efficiently station capacity is used.
- Many stations show **high utilization levels**, indicating strong usage.
- Some stations show **lower utilization**, meaning capacity is underused.
- High utilization stations indicate **popular locations**, while low utilization stations may need **relocation or demand improvement strategies**.

Bike availability by city



- **Lyon contributes the highest share of bike availability (around 50%)**, indicating the largest operational coverage.
- **Toulouse and Bruxelles-Capitale** also show significant bike availability compared to other cities.
- Smaller cities like **Seville, Dublin, and Marseille** contribute relatively lower availability.

Average Station Utilization (Gauge)



- The **average station utilization is around 46%**, which indicates moderate usage of station capacity.
- Utilization is **below the target level of 70%**, suggesting potential capacity underutilization.

Station Performance Table

Station Name	Sum of Occupied Stands	Average of Bik
富山事務所 Toyama Warehouse 2	2	
Wolfe Tone Street	0	
The Point	8	
Test Val	1	
South Dock Road	9	
Soseska Novo Brdo	2	
Smithfield North	5	
Šmartinski Park	11	
Savsko Naselje 1-Šmartinska Cesta	8	
Povšetova-Grablovičeva	18	
Povšetova - Kajuhova	0	
Oriel Street Test Terminal	1	
Total	**8937**	

- Some stations show **higher occupied stands**, indicating frequent usage.
- Several stations have **very low occupied stands**, highlighting uneven bike distribution across stations.

Map Visualization

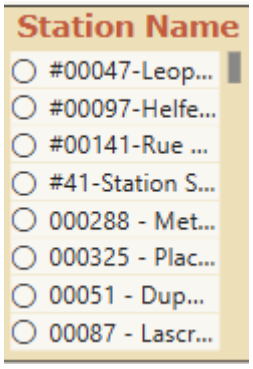
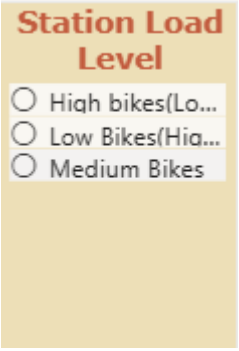
Map Visualization displays the geographic distribution of bike sharing stations using **latitude and longitude coordinates**. Each point on the map represents a bike station, allowing users to easily identify station locations and analyze the spatial distribution of bike availability across the city.

Trend analysis

How does total bike availability change across different years?

Strong Growth in 2025 Bike Availability
Total available bikes show a **sharp increase in 2025 compared to previous years**, indicating expansion in bike supply or improved station capacity to meet rising urban mobility demand.

2. **Stable Availability in Earlier Years**
From **2022 to 2024**, bike availability remained relatively **low and stable**, suggesting either limited station infrastructure or lower adoption of the bike-sharing service during that period

Slicers & Button Implemented

- **Station Name, Station Load Level, and Year slicers** are implemented to allow users to dynamically filter the dashboard and focus on specific stations, load categories, or time periods.
- These interactive filters help users quickly explore station performance, utilization patterns, and yearly trends within the bike sharing system.